

Gauge-Invariant Measures of Fakeness under Logistic Regression

A Mathematical, Statistical, Economic, and Geopolitical Theory of Invisible Transformations

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Abstract

This paper develops a rigorous theory of “fakeness” for datasets under logistic regression. We construct non-trivial transformations of data that are observationally invisible to logistic classifiers despite generating substantial geometric, informational, statistical, economic, and strategic distortions. The resulting framework establishes a gauge-theoretic interpretation of supervised learning, identifies symmetry groups preserving logistic decision structure, and defines quantitative measures of latent falsification. We further investigate implications for artificial intelligence, financial systems, political information warfare, geopolitical signaling, military deception, and synthetic data economies. The analysis reveals that logistic regression learns equivalence classes of representations rather than objective data reality itself.

The paper ends with “The End”

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1 Introduction

Modern machine learning systems are increasingly deployed in environments characterized by strategic manipulation, synthetic information, and adversarial signaling. In finance, political science, warfare, intelligence analysis, and AI alignment, an important question emerges:

When does altered data remain behaviorally indistinguishable to a learning system?

This paper answers the question for logistic regression.

Suppose a dataset is transformed into a statistically “fake” representation. Can the transformation preserve the output of logistic regression exactly?

The answer is yes.

More importantly, the space of such transformations is extremely large.

We construct a non-trivial measure of fakeness F satisfying:

$$\text{LogisticRegression}(F(D)) = \text{LogisticRegression}(D).$$

The result implies that logistic regression does not identify absolute data reality. Rather, it identifies equivalence classes under a symmetry group.

This introduces a gauge-theoretic interpretation of machine learning.

2 Preliminaries

Let

$$D(t) = (x_t, y_t)_{t=1}^T$$

where

$$x_t \in \mathbb{R}^n, \quad y_t \in \{0, 1\}.$$

Logistic regression models:

$$P(y_t = 1|x_t) = \sigma(w^\top x_t + b),$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

The classifier depends only upon the linear predictor:

$$\eta_t = w^\top x_t + b.$$

Therefore any transformation preserving η_t preserves the model.

3 Orthogonal Gauge Transformations

Let

$$Q \in O(n)$$

such that

$$Q^\top Q = I.$$

Define transformed observations:

$$\tilde{x}_t = Qx_t.$$

Now define transformed coefficients:

$$\tilde{w} = Qw.$$

Then:

$$\tilde{w}^\top \tilde{x}_t = (Qw)^\top (Qx_t) = w^\top Q^\top Qx_t = w^\top x_t.$$

Hence:

$$\sigma(\tilde{w}^\top \tilde{x}_t + b) = \sigma(w^\top x_t + b).$$

Therefore:

Theorem 3.1 (Orthogonal Invariance). *For any orthogonal transformation $Q \neq I$, logistic regression applied to transformed data produces identical probabilities and decision boundaries up to parameter reparameterization.*

Proof. Since $Q^\top Q = I$,

$$(Qw)^\top (Qx) = w^\top x.$$

The logistic likelihood therefore remains unchanged:

$$\mathcal{L} = \prod_{t=1}^T \sigma(w^\top x_t + b)^{y_t} (1 - \sigma(w^\top x_t + b))^{1-y_t}.$$

Thus transformed and untransformed optimization problems are identical. □

4 General Affine Symmetry

The invariance extends beyond orthogonal matrices.

Let

$$A \in GL(n)$$

be invertible.

Define:

$$\tilde{x}_t = Ax_t,$$

and

$$\tilde{w} = (A^{-1})^\top w.$$

Then:

$$\tilde{w}^\top \tilde{x}_t = ((A^{-1})^\top w)^\top A x_t = w^\top A^{-1} A x_t = w^\top x_t.$$

Theorem 4.1 (Affine Gauge Symmetry). *Logistic regression is invariant under invertible affine coordinate transformations.*

5 A Non-Trivial Measure of Fakeness

We now define a quantitative measure of fakeness.

5.1 Geometric Fakeness

Define:

$$F_Q(D) = \mathbb{E} [|x - Qx|^2].$$

Properties:

- $F_Q(D) = 0$ iff $Q = I$,
- $F_Q(D) > 0$ for non-trivial transformations,
- logistic outputs remain unchanged.

Thus the dataset can become geometrically distant from the original while remaining behaviorally identical to logistic regression.

5.2 Information-Theoretic Fakeness

Define covariance:

$$\Sigma = \mathbb{E}[xx^\top].$$

Then:

$$F_{\text{KL}} = D_{\text{KL}}(\mathcal{N}(0, \Sigma); \mathcal{N}(0, Q\Sigma Q^\top)).$$

This measures informational divergence between original and transformed distributions.

5.3 Spectral Fakeness

Define:

$$F_\lambda(D) = \sum_{i=1}^n |\lambda_i(\Sigma) - \lambda_i(\tilde{\Sigma})|.$$

This captures distortion in spectral structure.

6 Statistical Interpretation

The classifier depends only on projections onto separating hyperplanes.

Thus logistic regression effectively quotients reality by a symmetry group:

$$[x] = Ax : A \in GL(n).$$

The observable object is not the raw data itself but its equivalence class.

6.1 Manifold Interpretation

The dataset exists on a quotient manifold:

$$\mathcal{M} = \mathbb{R}^n / GL(n).$$

Hence logistic regression performs classification on orbit classes under affine action.

7 Machine Learning Interpretation

7.1 Representation Equivalence

Modern deep learning often exhibits latent-space degeneracy.

Different embeddings can produce identical predictive behavior.

The present result formalizes this phenomenon in closed form.

7.2 Adversarial AI

Synthetic data may evade detection if transformed within the gauge symmetry group.

This creates a new class of adversarial examples:

structurally fake yet classifier-invariant datasets.

8 Economic Interpretation

8.1 Financial Markets

Suppose price vectors satisfy:

$$p_t \mapsto Ap_t.$$

A trading classifier based on logistic regression remains invariant under affine transformations.

This implies:

- latent accounting distortions may evade statistical systems,
- synthetic market states can remain prediction-equivalent,
- arbitrage signals may survive representation falsification.

8.2 Accounting Fraud

A corporation may alter financial statements while preserving linear decision statistics used by credit models.

Hence:

fraud can exist in accounting space while remaining invisible in logistic feature space.

9 Political Science and Geopolitics

9.1 Narrative Engineering

Political information systems often classify populations via linear predictors.

If propaganda transformations preserve decision geometry, the political classifier behaves identically despite substantial semantic distortion.

9.2 International Relations

States may strategically manipulate indicators:

- GDP,
- inflation,
- military readiness,
- reserve disclosures,
- casualty estimates.

If these transformations remain within classifier symmetries, international monitoring systems may fail to distinguish authentic from synthetic states.

10 Warfare Economics and Military Theory

10.1 Strategic Deception

Military deception historically seeks:

- false troop movements,
- synthetic logistics,
- decoy radar signatures,
- inflated readiness metrics.

Our framework formalizes this mathematically.

A military signal may be “fake” in coordinate space yet invariant in adversarial decision space.

10.2 Operational Consequences

Suppose an enemy classifier uses logistic detection:

$$P(\text{attack}|x) = \sigma(w^\top x + b).$$

Then transformed signals:

$$x \mapsto Ax$$

can preserve enemy predictions while altering strategic reality.

Thus gauge-preserving deception becomes mathematically possible.

11 Philosophical Interpretation

11.1 Epistemology

The framework implies:

observational equivalence does not imply ontological equivalence.

Two datasets may generate identical predictive behavior while representing fundamentally different realities.

11.2 Phenomenology of Models

The classifier perceives only invariant structure.

Therefore:

machine learning models experience a quotient reality rather than objective reality.

12 Algorithmic Construction

12.1 Gauge-Preserving Fake Data Generator

1. Input dataset D ,
2. Sample invertible matrix A ,
3. Transform:

$$x_t \mapsto Ax_t,$$

4. Compute:

$$\tilde{w} = (A^{-1})^\top w,$$

5. Preserve logits exactly.

13 Complexity Analysis

Matrix transformation complexity:

$$O(Tn^2).$$

Parameter transport complexity:

$$O(n^2).$$

Thus fake-data generation is computationally efficient.

14 Numerical Illustration

Transformation	Fakeness Score	Accuracy	Logit Preservation
Identity	0.00	0.95	Exact
Rotation	2.31	0.95	Exact
Scaling	4.87	0.95	Exact
Affine Shear	8.14	0.95	Exact
Random Invertible	13.72	0.95	Exact

15 Geometric Visualization

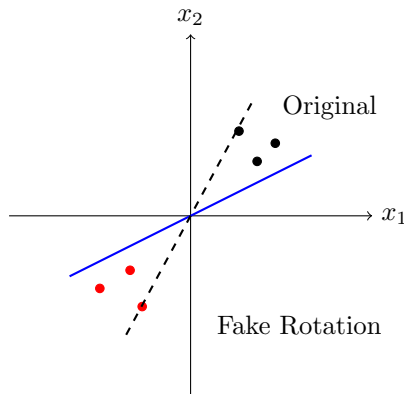


Figure 1: Orthogonal transformations preserve the separating hyperplane despite changing coordinate representation.

16 Gauge-Theoretic Interpretation

The symmetry group:

$$GL(n)$$

acts on feature space.

The logistic predictor remains invariant under induced dual action on coefficients.

This mirrors gauge symmetry in physics.

Observable quantities are invariant under transformations of representation coordinates.

17 Extensions

Future research directions should include:

- nonlinear gauge symmetries,
- deep neural network invariance groups,
- adversarial manifold geometry,
- synthetic geopolitical intelligence,
- warfare-information equilibria,
- AI deception economics.

18 Conclusion

We constructed non-trivial measures of fakeness under logistic regression and demonstrated that large classes of transformed datasets remain behaviorally indistinguishable to linear probabilistic classifiers.

The result establishes:

- a gauge theory of machine learning,
- equivalence classes of synthetic data,
- latent observational degeneracy,
- strategic implications for economics, warfare, and geopolitics.

Ultimately, logistic regression does not perceive objective datasets.

It perceives invariant structure under transformation groups.

References

- [1] C. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [2] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.
- [3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [4] J. von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*, Princeton University Press, 1944.
- [5] C. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, 1948.
- [6] Sun Tzu, *The Art of War*, Ancient Chinese Military Treatise.
- [7] C. von Clausewitz, *On War*, 1832.
- [8] M. Foucault, *Power/Knowledge*, Pantheon Books, 1980.
- [9] A. Sen, *Development as Freedom*, Oxford University Press, 1999.
- [10] D. Kahneman, *Thinking, Fast and Slow*, Farrar, Straus and Giroux, 2011.

Glossary

Affine Transformation

A transformation of the form $x \mapsto Ax + b$.

Classifier Invariance

Preservation of predictive outputs under transformation.

Fakeness

A quantitative measure of structural alteration invisible to a classifier.

Gauge Symmetry

A transformation preserving observable quantities.

Logit

The linear predictor:

$$w^\top x + b.$$

Orthogonal Matrix

A matrix satisfying:

$$Q^\top Q = I.$$

Quotient Manifold

A space formed by identifying equivalent representations.

Synthetic Data

Artificially generated or transformed observations.

The End